

# Jeff Shen

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## Education

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**H.B.Sc.**, Statistics, Astronomy & Astrophysics, Mathematics 2018-2022  
University of Toronto

## Technical Skills

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**Data:** Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Stan, Seaborn, SQL, NoSQL, R, Excel

**Web:** HTML5, CSS3, JavaScript, React (Native), Redux, Node, Express.js, Flask

**Other:** Rust, Docker, shell scripting, vim, version control (git), cloud computing (AWS, Azure, Google Cloud Platform), Unix/Linux, SymPy, Astropy, Selenium, NLTK,  $\text{\LaTeX}$ , regular expressions

## Research Experience

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| 2020 Aug. - Present  | <b>Undergraduate Research Course</b><br><i>Canadian Institute for Theoretical Astrophysics</i><br>Supervised by Prof. <a href="#">Gwendolyn Eadie</a> and Prof. <a href="#">Norman Murray</a> . Study of mass distribution of the Milky Way in a Bayesian framework.  |
| 2020 May - 2020 Aug. | <b>Summer Undergraduate Research Program</b><br><i>Dunlap Institute for Astronomy and Astrophysics, University of Toronto</i><br>Supervised by Dr. <a href="#">Allison Man</a> . Study of molecular gas in a strongly-lensed protocluster system at $z \approx 2.9$ . |

## Honours and Awards

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| 2020 | <b>Summer Undergraduate Research Program Award (\$9,500 CAD)</b><br>Dunlap Institute for Astronomy and Astrophysics, University of Toronto |
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## Independent Coursework

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| 2020 | <b>Practical Time Series Analysis</b> , The State University of New York (Coursera)<br>Stationarity, differencing, ACF, PACF, autocovariance. MA processes, AR processes, AR-MA duality. Backshift operator, difference equations, Yule-Walker matrix equations. AIC, SSE, model selection. ARMA and ARIMA models. Seasonality, SARIMA. Exponential smoothing, forecasting. |
| 2019 | <b>Data-Driven Astronomy</b> , The University of Sydney (Coursera)<br>Pulsars, black holes, exoplanets, stellar evolution, distances, redshift, galaxy classification. Image stacking, algorithm analysis, crossmatching, databases, regression, classification, machine learning. Implementation in Python with Astropy; SQL.                                              |
| 2019 | <b>Intro to Deep Learning with PyTorch</b> , Udacity<br>Mathematics of neural networks. Data preparation, regularization techniques. Image classification with CNNs and OpenCV. Time series with RNNs. Implementation in Python with PyTorch.                                                                                                                               |
| 2018 | <b>Intro to Machine Learning</b> , Udacity<br>Classifiers (naïve Bayes, SVMs, ensemble methods), data processing, feature selection, regression, clustering, outlier detection, PCA, evaluation metrics. Implementation in Python with scikit-learn.                                                                                                                        |

## Outreach

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| 2019 - Present | <b>Space Science Journalist (Current in Space segment)</b> , <i>The Star Spot</i> podcast<br>Curate, write and record stories about the latest research and news in astronomy.<br>Episodes 177 - present. |
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## Volunteering

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| 2015 - 2018 | <b>Python Workshop Lead</b> , Sir Winston Churchill Secondary, Vancouver, B.C.<br>Coordinated and delivered workshops to prepare students for national programming contest. |
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## Relevant Coursework

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|-------------|------------------------------------------|
| 2020 Fall   | Real Analysis                            |
| 2020 Fall   | Regression Analysis                      |
| 2020 Fall   | Practical Astronomy                      |
| 2020 Winter | Galaxies and Cosmology                   |
| 2020 Winter | Practical Physics I                      |
| 2020 Winter | Electricity and Magnetism                |
| 2020 Winter | Thermodynamics and Statistical Mechanics |
| 2019 - 2020 | Probability and Statistics I, II         |
| 2019 Fall   | Stars and Planets                        |
| 2019 Fall   | Ordinary Differential Equations          |
| 2019 Summer | Multivariable Calculus                   |
| 2019 Winter | Computer Science                         |
| 2018 - 2019 | Calculus                                 |
| 2018 - 2019 | Linear Algebra I, II                     |
| 2018 Fall   | Statistical Reasoning and Data Science   |

## Interests

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**Languages:** English (native), French, Chinese, Norwegian (basic)

**Other:** \*nix customization, computer graphics, 3D modelling, running, baking, reading, guitar